SAMYAK JAIN

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Professional Summary

Software Engineer experienced in building backend services, event-driven systems, frontend applications, shared developer tooling, and cloud infrastructure for a multi-tenant K–12 EdTech platform. Strong foundation in API design, distributed systems, data modeling, object-oriented programming, and data structures and algorithms.

Experience

Compro Technologies — Delhi *Jan 2025 – Present*

Software Engineer

Project: K–12 LMS (School Framework) — multi-tenant platform supporting 1M+ active users across schools

- Built the core Roster Service to model **6 OneRoster entities** -- organizations, users, courses, classes, enrollments, and academic sessions -- per the 1EdTech OneRoster data standard, with dedicated Lambda handlers per entity to process incoming roster data.
- Integrated Rostersteam (Ednition) as the roster ingestion source, consuming OneRoster-formatted CSV data through event-driven Lambda handlers to **automate onboarding** across multiple schools.
- Built a Calendar Service end to end -- event modeling, ingestion, and calendar-date processing following the EdFi standard schema -- using a **CQRS-based read model** for user-course aggregations and progress tracking, improving read throughput by roughly **35%** while supporting multi-tenancy.
- Architected and delivered an **event-driven transcript generation platform** using AWS SQS, Lambda, DynamoDB, API Gateway, and S3, maintaining **99.9% availability**.
- Reduced frontend load times by cutting **FCP and LCP by ~0.5s** via image optimization, lazy loading, and code splitting; and refactored **15+ legacy modules** into a reusable component library, improving maintainability across shared applications.
- Fixed a **cross-tenant session-leakage vulnerability** across the Onboarding and Teaching applications, restoring correct tenant, session, and domain isolation.
- Built and documented an internal service SDK and provisioned infrastructure via AWS CDK across **5 microservices**.
- Shipped **15+ REST APIs and 10+ GraphQL APIs** with JWT authentication, request validation, caching, rate limiting, centralized structured logging, and automated error handling -- reducing incident-investigation time by roughly **30%**.

Projects

Linkora -- Scalable URL Shortener *Nuxt.js, Vue.js, TypeScript, Express.js, DynamoDB, AWS CDK*

Live: linkora-mnyr.onrender.com | **Code:** github.com/imsamyak/url-shortner | **Architecture:** linkora-mnyr.onrender.com/architecture

- Engineered a **full-stack URL shortener** with a **Nuxt 4/Vue 3 SSR client** and a modular **Express 5/TypeScript REST API**; implemented authenticated workspaces, link creation, expiration, and redirection.
- Designed a **single-table DynamoDB model** with **cursor-based pagination** and **native TTL attributes** for scalable access patterns and automatic cleanup of expired redirects.
- Secured API flows with **password hashing**, **intent-scoped authentication**, **schema validation**, and **protected server-side gateways**.
- Provisioned a **two-AZ cloud architecture** with traffic distribution, auto scaling, firewall protection, and monitoring, plus rolling CI /CD; tested core services with unit and end-to-end tests.

Education

Maharaja Agrasen Institute of Technology, GGSIP University — Delhi

Bachelor of Technology in Computer Science and Engineering — CGPA: 8.95 *2025*

Technical Skills

Languages: JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frontend & UI: React.js, Tailwind CSS, SCSS

Backend & APIs: Node.js, Express.js, NestJS, REST, OpenAPI, GraphQL

Databases: MySQL, PostgreSQL, DynamoDB

Cloud & DevOps: AWS Lambda, AWS S3, AWS SQS, AWS SNS, CloudWatch, Docker, CI/CD

Tools & Testing: Git, GitHub, Postman, SonarQube, Vitest, Playwright

Concepts: Data Structures & Algorithms, SOLID Principles, API Design, Agile Methodologies, OOPs, Microservices, Event-Driven Architecture, Domain-Driven Design, CQRS, Cloud Computing, Core Web Vitals

AI-Assisted Development: GitHub Copilot, Claude